Presented at the American College of Sports Medicine Annual Meeting 2026

Development Of A Sensitive And Specific Portable Eye-tracking Diagnostic Tool For Concussion

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ABSTRACT

Objective eye movement metrics have been demonstrated to be useful in distinguishing concussed from uninjured individuals, but requires bulky equipment that cannot be used in far-forward military settings.

PURPOSE

To develop a rapid portable eye-tracking tool sensitive and specific for the diagnosis of concussion

METHODS

We recruited 189 concussed (35.4% Female) and 1314 uninjured participants (18.7% Female) ages 17-25 years (mean 19.0, SD 1.1) from three sites (United States Military Academy at West Point, Children's Hospital of Philadelphia, University of Wisconsin - Green Bay). Eye movement metrics and symptom severity with eye movement were collected from uninjured individuals as a pre-injury baseline and from concussed individuals within 72 hours of injury. Objective eye movement metrics were obtained while participants viewed a 90-second video traveling around the periphery of a visual display. Size and position of pupils were recorded at 167 Hz with an infrared eye-tracker. Participants also rated symptom severity during eye movement measurement. The sample reserved for model development (116 concussed, 998 uninjured) was bootstrap resampled for a logistic regression model predicting injury status. For regression, eye movement metrics were expressed as one composite predictor and symptom severity was expressed as a second predictor, both in standardized units.

RESULTS

Measure	Concussed (n=189)	Uninjured (n=1,314)
Symptom Severity Score	26.3 (SD 16.5)	2.61 (SD 5.3)
Eye Movement Score	6.3 (SD 3.1)	4.0 (SD 2.7)

Average symptom severity score was 26.3 (SD 16.5) for injured participants vs. 2.61 (SD 5.3) for uninjured participants. Average eye movement score for injured participants was 6.3 (SD 3.1) vs. 4.0 (SD 2.7) for uninjured participants. The model based on both eye movement metrics and symptom severity with eye movement was 92.2% sensitive (95% confidence interval (CI) 85.8 - 96.4%), 93.1% specific (95% CI 91.3 -94.6%) in diagnosing concussion. A one-standard-deviation increase in standardized symptom severity resulted in a 5.3% (95% CI 3.1 - 7.6%) increase in concussion probability while a one-standard-deviation increase in the number of disrupted eye movement metrics resulted in a 3.1% (95% CI 1.4 - 5.1%) increase. Coefficients were positive in 100% of bootstrap runs.

CONCLUSION

A rapid portable eye-tracking device is highly sensitive and specific for the diagnosis of concussion with a test that is under two minutes.